Field Hours

CSCE 753 Computer Vision and Robot Perception

Original notes for this site. Same topics as the catalog course. Not a reprint of another university's lectures, and not academic credit.

Lecture 1 · 90 minutes

The projective camera

A pinhole camera sends a point (X, Y, Z) to a pixel. With focal length f and principal point (cx, cy), you set u to f times X divided by Z, plus cx, and you do the same for Y. Take f = 500, principal point (320, 240), and the point (1, 0, 5). Then u is 420 and v is 240. Division by Z is why equal steps look smaller when they are farther away.

You keep the ray and lose the distance. The points (1, 0, 5) and (2, 0, 10) share the ratio 0.2, so both hit one pixel. One image cannot tell 5 meters from 10 meters. You need a second view, a known size, or a range sensor. On a 640 by 480 image the principal point is near (320, 240) only if the sensor is centered. You measure that point rather than assuming it.

Pinhole projection drops the depth.

Divide X and Y by Z -> Scale by the focal length -> Add the principal point -> Keep the pixel, lose the range

Two points, one pixel

1. Focal length 500 and principal point (320, 240). The point (1, 0, 5) projects to u = 100 plus 320, so 420, and v = 240.

2. The point (2, 0, 10) has X divided by Z equal to 0.2 as well, so it projects to the same pixel (420, 240).

3. The depth ratio is 2, and the lateral coordinate also doubled. That is why the pixel did not move. A single view recorded the ray, not a unique metric point.

Problem 1. Where does (1, 0, 5) project, with focal length 500 and principal point (320, 240)?

Answer. (420, 240). The horizontal coordinate is 500 times 1 divided by 5, plus 320.

Problem 2. Why can this image not decide between (1, 0, 5) and (2, 0, 10)?

Answer. Both points have X divided by Z equal to 0.2 and Y equal to 0, so both project to (420, 240). One view stores the ray, not a unique depth.

Lecture 2 · 90 minutes

Two views

Two cameras and a point give two rays that are coplanar with the baseline. After rectification the match lies on the same row. If one camera sees u = 420 and the other, shifted by 0.1 meters, sees u = 410, the disparity is 10 pixels. With focal length 500 the depth is 0.1 times 500 divided by 10, so 5 meters. You search that row, not all

640 by 480 pixels.

The essential matrix stores rotation and translation direction, not metric scale. Baselines (0.1, 0, 0) and (0.2, 0, 0) give the same direction, because scaling the scene with the baseline does not change rays. You recover meters from a known length. Five correspondences are the calibrated minimum, and eight are the classical count for the fundamental matrix. Once matches are wrong, you need more than that minimum.

What a second camera adds.

One view

One ray per pixel

Depth stays free

Focal length and principal point

Two views

Match on the epipolar line

Disparity over a known baseline

Depth from the two rays

A 10 pixel shift along the row

1. Baseline 0.1 meters, focal length 500, disparity 10. Depth is 50 divided by 10, so 5 meters.

2. If the second match is actually at a disparity of 20, the depth is 50 divided by 20, so 2.5 meters.

3. A 10 pixel mistake along the row changed the depth by a factor of 2. The epipolar constraint told you which row to search. It did not guarantee the pixel you picked on that row.

Problem 1. Baseline 0.1 meters, focal length 500, disparity 10. What is the depth?

Answer. 5 meters. The product of baseline and focal length is 50, and 50 divided by 10 is 5.

Problem 2. Why do the translations (0.1, 0, 0) and (0.2, 0, 0) give the same essential-matrix direction?

Answer. They are the same direction at two scales. Image rays do not change if you scale the scene and the baseline together, so the essential matrix does not store meters. You need a measured baseline or another known length.

Lecture 3 · 90 minutes

Estimating the camera

A 3 by 4 camera matrix has 12 entries up to scale, hence 11 degrees of freedom. Each 3D-to-2D correspondence gives you 2 equations, so 6 points are the classical minimum. Those 6 can be interpolated when there is no noise. With 20 points you solve a least-squares system. A mean residual of 0.3 pixels on the fitting points measures the fit, not a new view.

You then split the matrix into intrinsics and extrinsics. Focal length should land near the number you expect, for example 500 pixels. A result of 50 pixels on an image 640 wide usually means the 3D units disagree with the

point list. Radial distortion adds 2 coefficients, because a pure pinhole model leaves error that grows toward the corner. Residuals of 0.3 pixels at the center and 4 at the corner are that pattern.

Fit the matrix, then read the residual.

1. Collect at least 6 known points

2. Solve the linear camera matrix

3. Split intrinsics and extrinsics

4. Read the corner residual

Center versus corner

1. Six noiseless points can fix the 11-degree-of-freedom matrix. You add 14 more, for 20 points in all.

2. The mean residual on the 20 is 0.4 pixels. Center points sit under 0.3 pixels. One corner point sits at 3 pixels.

3. A two-coefficient distortion model brings that corner residual down to 0.5 pixels. The linear pinhole fit was not a broken solver. It was the wrong physical model at the edge of a 640-pixel image.

Problem 1. How many known correspondences are the minimum for this 11-degree-of-freedom camera, and why is a zero residual on exactly that many not a confirmation?

Answer. Six. Each point supplies 2 equations, and six points cover the 11 degrees of freedom together with the homogeneous scale. A zero residual on those six points means the solver interpolated them. A seventh point is the check.

Problem 2. The center residual is 0.3 pixels, the corner residual is 4 pixels, and the focal length comes out near 500. What should you add?

Answer. A radial distortion model, typically two coefficients. The focal length is already near 500, and the error grows toward the corner rather than sitting everywhere. A new focal length is not the first change.

Lecture 4 · 90 minutes

Features and matches

A detector picks repeatable points and a descriptor summarizes a window so you can match across views. A 16 by 16 window is 256 numbers. You compare by Euclidean distance, or by Hamming distance if each entry is a bit. Accept a match at nearest distance 20 and second distance 50, ratio 0.4, when your threshold is 0.8. The second neighbor stops a wall of identical bricks from matching every brick.

A learned descriptor can replace that summary by 128 numbers, trained so a true match beats a false one by a margin such as 0.5. Your later tests do not change. You still threshold a ratio and still reject a match off the epipolar line. From 1000 detections you might keep 150 after the ratio test and 80 after the geometric check. The descriptor did not certify those 80. The geometry check did.

From a window to an inlier.

Detect a repeatable point -> Build the descriptor -> Apply the ratio test -> Drop matches off the epipolar line

Two ratios

1. Nearest distance 20 and second nearest 50 give the ratio 20 divided by 50, so 0.4. A threshold of 0.8 accepts the match.

2. Another point has distances 30 and 36. The ratio is 30 divided by 36, so 0.833, which fails 0.8. You discard it even though 30 is only modestly larger than 20.

3. Of 150 matches that pass the ratio test, a single geometric model keeps 80 and drops 70. Those 70 agreed with a local distance test and disagreed with one global motion.

Problem 1. Nearest distance 20, second nearest 50, threshold 0.8. Do you accept the match, and what is the ratio?

Answer. Yes. The ratio is 0.4, which is below 0.8.

Problem 2. Why can a nearest distance of 30 be rejected while a nearest distance of 20 is kept, aside from 30 being larger?

Answer. The test uses the ratio to the second nearest, not the raw distance alone. 30 divided by 36 is about 0.83 and fails 0.8, while 20 divided by 50 is 0.4 and passes. Repeated texture produces a small distance to several wrong windows, and the ratio is what catches that.

Lecture 5 · 90 minutes

Consensus under bad matches

You fit a model on a minimal subset and count how many points agree. A line needs 2 points. With 100 correspondences and inlier rate 0.5, one draw of 2 points is clean with probability 0.25. The chance a draw is dirty is 0.75. After 10 draws, the chance of no clean pair is about 0.056, so about 0.944 that at least one clean pair showed up.

You then refit on every point inside the threshold, not only on the 2 that proposed the model. A gate of 2 pixels keeps a point whose error is at most 2. A gate of 20 pixels will treat outliers as inliers and pull the refit. If the best sample holds 60 of 100 points, you refit those 60. The fraction you saw is 0.6, which can beat the 0.5 you assumed.

One consensus loop.

1. Draw a minimal subset

2. Fit the model

3. Count points within 2 pixels

4. Refit the largest consensus

Ten draws and a refit

1. Inlier probability per point is 0.5 and the sample size is 2, so one draw is clean with probability 0.25.

2. The probability that all 10 draws are contaminated is about 0.056. The probability that at least one is clean is about 0.944.

3. The best consensus holds 60 points. You refit on those 60. You do not refit on all 100, because the other 40 missed the 2 pixel test and would pull the model.

Problem 1. If each point is an inlier with probability 0.5, what is the probability one draw of 2 points is clean, and about what is the probability that 10 draws include at least one clean draw?

Answer. 0.25 for one draw. About 0.944 for ten draws, because the probability of missing every time is about 0.75 to the tenth, which is about 0.056.

Problem 2. Why do you refit on the 60 inliers rather than on the original 2 points or on all 100?

Answer. The original 2 are only a minimal sample. All 100 include the 40 points outside the 2 pixel threshold, and those points pull the model. The refit uses the consensus set the threshold defined.

Lecture 6 · 90 minutes

Lidar points

A spinning lidar returns range at a known angle. A beam at 10 meters with sigma 0.02 meters is a 3D point once you apply the sensor frame. One turn with 1024 azimuth slots and 32 rows has 32768 possible returns. You drop max-range misses, often beyond 100 meters, before building a cloud. A max-range reading is not a surface you should keep.

Point-to-plane registration pairs each point with a plane and cuts the distance to it. A 0.1 meter error along x can fall in one iteration when some plane constrains x. A corridor of parallel vertical planes does not constrain motion along the hall. That along-track error can stay near 0.1 meters for as many iterations as you run. You need geometry that breaks the symmetry, or a second sensor.

From beam slots to surface points.

Angle plus a range

Drop max-range misses

Transform into the world

Keep the surface returns

A corridor that does not lock

1. There are 32768 beam slots, and 2000 of them are max-range. You keep 30768 surface returns.

2. In a room with a floor and an end wall, point-to-plane pairs can pull an initial 0.1 meter error along x down, because the end wall constrains that axis.

3. In a corridor with no end wall in view, the same iteration leaves the along-track error near 0.1 meters. The cost can still fall a little by sliding along the free direction. A falling cost is not a fully constrained solution.

Problem 1. How many surface returns do you keep if 2000 of 32768 slots are max-range?

Answer. 30768.

Problem 2. Why can the point-to-plane cost fall in a corridor while the along-track error stays near 0.1 meters?

Answer. The visible planes do not constrain motion along the corridor. Sliding on that axis barely changes point-to-plane distance, so the cost can look fine while that coordinate is still wrong by about 0.1 meters. You need geometry that breaks the symmetry.

Lecture 7 · 90 minutes

Visual and lidar SLAM

Visual SLAM tracks features and estimates poses and points together. You keep a keyframe after motion of about 0.2 meters or 10 degrees, so you skip a near-duplicate view. Bundle adjustment sums squared reprojection error. A miss of 3 pixels contributes 9. If 200 keyframes are too slow, you optimize a window of the last 5.

Lidar SLAM uses a point-to-plane residual in meters and adds a pose-graph edge when a loop closes. A 0.4 meter loop residual is what makes 0.4 meters of drift observable. Low texture can drop your visual inliers from 80 to 5, and a bare corridor can leave lidar free along one axis. You drop the modality whose count collapsed. Averaging a 5-inlier pose with a good one gives the weak pose a vote it did not earn.

Two residuals, two units.

Visual SLAM

Feature tracks across frames

Squared error in pixels

A keyframe every 0.2 meters

Lidar SLAM

Point-to-plane pairs

Residual in meters

Loop edges on the pose graph

Pixels and meters stay separate

1. Ten points, each 3 pixels off, contribute 10 times 9, so 90, to the visual sum of squares. After adjustment, if each error is 1 pixel, the contribution is 10.

2. A lidar loop residual of 0.4 meters is a different unit. You add it as a pose-graph edge. You do not add 0.4 to the pixel sum without a scale you declared.

3. On the newest frame the visual inlier count is 5, down from 80. You drop the visual relative-pose edge for that frame and keep the lidar edge.

Problem 1. Ten points each have a reprojection error of 3 pixels. What do they add to the sum of squares, and what do they add after each error falls to 1 pixel?

Answer. 90 before, because 10 times 9 is 90. Afterward the contribution is 10, because each squared error is 1.

Problem 2. Why is averaging a visual pose fit to 5 inliers with a lidar pose a poor fusion rule here?

Answer. The visual pose was fit to 5 matches, down from a typical 80, so that front end is starved. An average gives it the same vote as a lidar pose that still has geometry. The rule is to drop the modality whose inlier count collapsed.

Lecture 8 · 90 minutes

Residuals you can name

You can project a point with a focal length and a principal point, and you can say why one view leaves a ray. You can recover depth from disparity and a baseline, count 11 degrees of freedom in a camera matrix, and accept a match with a ratio threshold of 0.8. You can draw a minimal sample, keep on the order of 30000 lidar returns, and name a residual in pixels or in meters.

You can put two-view depth and a mapping residual in one argument. Baseline 0.1 meters and disparity 10 pixels pin a depth of 5 meters, only as good as inliers inside a 2 pixel gate. A pixel sum of squares will ignore a lidar axis the planes do not lock, unless you ask which coordinate is actually constrained. The system you can audit is a projection, a set of inliers, and a residual whose units you can name.

Audit a metric point before you use it.

1. Name the projection model

2. Keep the inliers inside the gate

3. Attach a residual with units

4. Refuse an unconstrained axis

A disparity gate is a depth gap

1. Baseline 0.1 meters and focal length 500 give the product 50. Disparity 10 means depth 5 meters. That match was kept by a consensus gate of 2 pixels.

2. A disparity of 12, still inside a slip of 2 pixels, gives depth 50 divided by 12, about 4.17 meters. The difference from 5 is about 0.83 meters.

3. You should report that depth gap with the 5 meters. The image ray from the first lecture still cannot separate two points on its own. A lidar range along the same ray is what supplies the meters.

Problem 1. Combine the two-view depth formula with the consensus gate. Baseline 0.1 meters, focal length 500, disparity 10, and a 2 pixel gate. What depth does disparity 10 give, and what depth does disparity 12 give?

Answer. Disparity 10 gives 5 meters. Disparity 12 gives 50 divided by 12, about 4.17 meters. The gate that accepted the match already allows a depth shift of about 0.83 meters.

Problem 2. Combine one-view projection with a lidar range. The image cannot separate (1, 0, 5) from (2, 0, 10), and a lidar return along that ray reads 5.1 meters. What metric point do you report, and what

did the image still supply?

Answer. You place the point on the camera ray at the lidar range. The ray has X divided by Z equal to 0.2, so at 5.1 meters the point is about (1.02, 0, 5.1). The image supplied the ray direction. The lidar supplied the metric depth one view did not have.